

Date	15/07/2026
Team ID	XXXXXX
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	5 Marks

Solution Architecture Template

Technical architectural flow and component details.

Three-Tier Architectural Layout

1. **Presentation Layer (UI):** Built with HTML5 and custom Vanilla CSS. Renders home, predict, and submit template pages inside a web browser.
2. **Application Layer (Server):** The Flask web application server (app.py) handles routing and processes POST form payloads.
3. **Core Prediction Layer (ML):** Loads the serialized Random Forest model and StandardScaler. Normalizes feature values and makes inference predictions.

Component Description Table

Component Name	Role in Solution	Technologies Used
Frontend UI	Displays form, collects applicant details, renders approval output.	HTML5, CSS3, Flask templates
Flask Controller	Handles HTTP endpoints (/predict, /submit), routes requests.	Python, Flask
StandardScaler	Scales numerical inputs (ApplicantIncome, CoapplicantIncome, LoanAmount, Term).	Scikit-learn (scaler.pkl)
Random Forest Model	Classifies applicant feature vector as approved (1) or rejected (0).	Scikit-learn (model.pkl)